SkillSignal Model Card

1. Model details

Task	Score how closely a resume matches a job description, then rank the resumes for that job
Model A	TF-IDF with unigrams and bigrams (sublinear tf, English stop words removed), compared by cosine similarity
Model B	all-MiniLM-L6-v2 sentence embeddings. Long texts are cut into 200-token chunks with a 50-token overlap, and the chunk vectors are averaged before taking cosine similarity
Real data	cnamuangtoun/resume-job-description-fit (Hugging Face), test split. We sampled 5 jobs with seed 42, giving 302 resume-job pairs. Labels: No Fit = 0, Potential Fit = 1, Good Fit = 2
Synthetic data	15 made-up Business Analyst profiles, each written once short (concise) and once wordy (verbose) with the same skills
Training	None. Neither model learns from the labels. We only use the labels to check the rankings

2. Intended use

- **Intended:** a first-pass skill match that a recruiter reads alongside the resume, with the matched and missing skills listed. It is also a class project that tests how much writing style moves the scores.
- **Out of scope:** rejecting or shortlisting people automatically, any hiring decision without a person checking it, and resumes not written in English. The scores are similarity values. They are not the chance that someone gets hired.

3. Overall performance (real data, 5 jobs)

Ranker	NDCG@5 [95% CI]	Precision@5	Recall@5
Embeddings (B)	0.561 [0.27, 0.84]	0.560	0.268
TF-IDF (A)	0.478 [0.17, 0.70]	0.480	0.258
Random order (average of 1,000 shuffles)	0.423 [0.30, 0.54]	0.412	0.150
Longest resume first	0.214 [0.06, 0.50]	0.240	0.095

The intervals all overlap. With only 5 jobs we cannot say which model ranks better. Intervals come from resampling jobs and then resumes 1,000 times.

4. Performance across text subgroups / styles

> **Main result.** In all 15 pairs, the concise version scored higher than the verbose version of the same profile, under both models. We also compared every concise profile against every verbose profile that had more skills. The concise one still won 75 of 75 times under TF-IDF and 67 of 75 times under embeddings.

(a) Concise vs verbose (synthetic)

15 profiles in 2 styles, scored against one Business Analyst job. All 30 texts were ranked together.

Style	Model	Mean score	Spearman: score vs real skill count	Mean rank (of 30)	Places in top 10
Concise	TF-IDF	0.154	0.68	7.8	10
Verbose	TF-IDF	0.020	0.23	22.8	0
Concise	Embeddings	0.422	0.22	8.7	9
Verbose	Embeddings	0.341	0.61	22.3	1

Our skill extractor found every skill in the concise texts but only 8 of 66 (12%) in the verbose ones. It also wrongly matched *stakeholder management* 7 times, from a stock sentence ending in "stakeholders", so its verbose precision is 53%.

(b) Short vs long real resumes

Split at the median length of 644 words. AUC is the chance that, for the same job, a relevant resume scores above a No Fit one.

Subgroup	Pairs (resumes)	% relevant	TF-IDF AUC [95% CI]	Embeddings AUC [95% CI]
Short (644 words or fewer)	155 (113)	46%	0.55 [0.45, 0.64]	0.55 [0.44, 0.64]
Long (over 644 words)	147 (112)	42%	0.54 [0.43, 0.64]	0.56 [0.46, 0.66]

Neither model separates relevant from irrelevant resumes well in either group, since every interval includes 0.5. Length does matter for TF-IDF, though. Long No Fit resumes sit 21 percentile points higher in their job's ranking than short No Fit resumes (95% CI 12 to 29; 85 long, 84 short). For embeddings the gap is 2 points (95% CI -7 to 11).

5. Explainability check

Method 1: top TF-IDF features

For each shared term we multiplied its weight in the resume by its weight in the job ad. These products add up to the TF-IDF score, so the biggest ones show what drove it. Profile_1 lists all 8 required skills in both versions.

Version	TF-IDF score	Shared terms	Terms that added the most
Concise	0.266	19	management process, gathering stakeholder, process, data, improvement, data analysis
Verbose	0.023	3	business, producing, strong

The verbose score is 91% lower, and none of its three shared words is a skill. Two of the concise version's top terms, "gathering stakeholder" and "management process", only exist because the skills are listed in the same order as in the job ad.

Method 2: embedding inspection

We embedded each skill name and its verbose wording, for example *excel* and "advanced spreadsheet modeling and reporting tools". A skill and its own rewording had an average cosine similarity of 0.44. A skill and a different skill's rewording averaged 0.19. Six of the 8 rewordings were closest to the right skill. The two weakest were *stakeholder management* (0.25) and *data analysis* (0.33). This is why the embedding score for Profile_1 only fell from 0.404 to 0.383. The concise texts still scored higher overall.

A real-data failure: one No Fit resume came 1st of 87 under TF-IDF and 2nd under embeddings. Its top TF-IDF terms were general words like "stakeholders" and "business processes", and it contains every skill our dictionary found in the job ad. Either the label is wrong or the models were fooled by shared generic wording. We have not checked which.

6. Limitations and recommendations

- **Main limitation: wording.** Both models reward how a resume is phrased more than what the person can do. Under TF-IDF, every verbose profile with all 8 skills ranked below every concise profile with only 3 (21 of 21 comparisons; 16 of 21 under embeddings). Nobody should use these scores to rank or filter candidates without reading the resumes.
- **Small samples.** We tested 5 real jobs, and 3 of them have no Good Fit resumes. The synthetic set is 15 pairs built from one template for one job. The intervals are wide, and the synthetic results only tell us about this experiment.
- **Labels not checked.** We did not hand-check the dataset's Fit labels, and the failure case above may be a labelling error.
- **Skill matching is basic.** It looks up exact words from a fixed list. It misses reworded skills (12% recall on verbose text) and picks up stock phrases.
- **Not tested.** We did not measure how hired people actually performed, or check for bias by gender, age, or other protected traits. This audit looks at writing style only.
- **Next steps.** Add common rewordings to the skill list, test real resumes rewritten by an LLM, and show the matched and missing skills next to every score.